

Audio Features from the Wav2Vec 2.0 Embeddings for the ACM Multimedia 2022 Stuttering Challenge

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ABSTRACT

The ACM Multimedia 2022 Stuttering Challenge is to determine the stuttering-related class of a speech segment. There are seven stuttering-related classes and an eighth garbage class. For this purpose, we have investigated the Wav2Vec 2.0 deep neural network to extract audio features from Transformer embeddings. Experiments were conducted on a part of the Kassel State of Fluency Corpus (KSoF). First, we introduced 21 functionals allowing to define two large composite audio feature: the first one set called W2V2 Basic audio-feature set (193,536 features) from the Base version of Transformer embeddings and the second one called W2V2 Large audio-feature set (516, 096 features) from the Large version of Transformer embeddings. Some functionals aim at estimating the spatial variability (e.g., mean, standard deviation, quartiles) and others aim at the temporal variability (e.g., linear regression slope). Then, each composite audio feature set have been splitted into a set of audio feature sets by grouping audio features by functional and by layer. Then, the most discriminant audio feature sets have been selected from these audio feature sets. Finally, two audio features sets specializing in stuttering speech have been developed and assessed: W2V2 Advanced audio-feature set (9,984 features) and W2V2 Large Advanced audio-feature set (15,360 features). Experiments have shown an improvement of 9.3% to the first one and 11.9% for the second one on the Test set compared to the official baseline of the Challenge (40.4%).

CCS CONCEPTS

- Information systems → Specialized information retrieval
- Computing methodologies → Feature selection

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KEYWORDS

Computational Paralinguistics; Stuttering Recognition; Audio Features; Transformer embeddings; Functionals; Challenge; Benchmark

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1 INTRODUCTION

Stuttering [1] is a multifactorial speech disorder defined by frequent prolongations, repetitions or broken words (e.g., pauses within a word). The most common form of stuttering begins during childhood. 80-90% of stuttering starts before the age of 6 and affects about 5% of children. Spontaneous recovery occurs in about 75% of individuals before adulthood. Stuttering has a male/female ratio of about 4:1. The vocal characteristics of stuttering are described by the American Psychiatric Association [2]. It is a disruption of the normal fluency and timing of speech, inappropriate for the individual's age. According to this table, stuttering speech is manifested by seven types of speech dysfunctions: Sound and Syllable repetitions, Broken words, Audible or Silent blocking, Circumlocution, Words produced with a physical tension, Monosyllabic whole-word repetitions. Stuttering may also be accompanied by inappropriate movements such as head shaking, eye blinking, lip or face trembling, and fist clenching

A vast amount of literature is available about the acoustic features and the classification techniques related to stuttered speech [3, 4, 5]. Usual speech analysis parameters were used: Mel Frequency Cepstrum Coefficients (MFCC), Linear Predictive Coding (LPC), Perceptual Linear Predictive (PLP), Formant, Pitch, Shimmer, Zero-Crossing Rate (ZCR) and Energy. These parameters allow the detection of stuttering speech with well-known classifiers in speech recognition: Hidden Markov Model (HMM), Support Vector Machine (SVM), K-Nearest Neighbors

(KNN). The classification tasks first focused on the major speech disorders: the repetitions considered most common in stuttering, prolongations and blocks [3, 4, 5]. For the detection of repetitions, speech recognition techniques such as HMM [6] or DTW [7] have been used. In [8] the syllable repetitions, syllable-initial prolongations, and blocks with stop consonant repetitions have been studied and detected using an Artificial Neural Networks (ANN) [8, 9]. The results reported in the survey [3] on Stuttered Speech Recognition shows that the detection of repetitions, prolongations and blocks gives an accuracy between 85% and 90%. In [10], six speech disfluencies related to stuttering (sound repetition, word repetition, phrase repetition, revision, fillers, prolongation) were detected independently of each other using bidirectional long short-term memory (Bi-LSTM). The same network architecture has been used [11] for 5-classes detection (word repetition, sound repetition, fillers, prolongation, non-stutter). Eight types of stuttering (sound repetition, part-word repetition, word repetition, prolongation, phrase repetition, fillers, revision, blocking) have been classified [12] using language model on the transcript of the speech [13]. Multi-classes detection (fluent, block, repetition, prolongation) using a Time Delay Neural Network (TDNN) with MFCCs [14], and Wav2Vec2.0 (W2V2) Embeddings is proposed in [15]. Many of these studies have been conducted on non-public dataset or on public dataset [16, 17] with non-public annotations. Two public datasets with public annotations are now available: Stuttering Events in Podcasts (SEP-28k) dataset [18], and Kassel State of Fluency Dataset [19] which is used for the Stuttering Challenge [20].

2 SPEECH MATERIAL

The Stuttering Challenge Corpus (KSF-C) consists in 4,601 three seconds speech chunks recorded by 37 German speakers. It originates from the Kassel State of Fluency (KSoF) corpus [19]. No textual transcript is provided and it appears that no two speech chunks have the same vocal content. The corpus was divided into three speaker independent partitions: a Train set (2,489 speech chunks), a Devel set (982 speech chunks) and a Test set (1,130 speech chunks). There are seven stuttering-related classes (block, fillers, modified speech technique, no disfluency, prolongation, sound repetition, word/phrase repetition) and an eighth garbage class. Table 1 gives the characteristics of KSF-C for the corpora Train and Devel.

Table 1: Characteristics of the Train and Devel sets

Class	Train		Devel	
	#Oc.	%Oc.	#Oc.	%Oc.
Block	310	12.5	102	10.4
Fillers	205	8.2	104	10.6
Garbage	52	2.1	33	3.4
Modified speech technique	687	27.6	185	18.8
No disfluency	830	33.3	444	45.2
Prolongation	183	7.4	53	5.4
Sound repetition	169	6.8	38	3.9
Word/Phrase repetition	53	2.1	23	2.3

We notice that the classes are very unbalanced with a different occurrence rate between the Train and the Devel sets. For instance, the occurrence rate for no disfluency is 33.3 % on Train set (45.2% on Devel set) while this occurrence rate for phrase repetition is 2.1 % on Train set (2.3% in Devel set), which is a ratio of 15.9 on Train set (19.7 in Devel set).

3 W2V2-EMBEDDING BASIC SYSTEM

Our goal is to obtain a single audio feature vector for the entire conversation regardless of its duration. We have used various functionals (global statistical functions) on Wav2Vec 2.0 (W2V2) [21] embeddings to obtain feature vectors of equal size for each speech utterance. Some functionals aim at estimating the spatial variability (e.g., mean, standard deviation, skewness, kurtosis, quartiles, interquartiles) and others aim at the temporal variability (e.g., linear regression slope, quadratic regression error). We were inspired by the approach used for the OpenSMILE toolkit [22] by replacing the audio Low-Level Descriptors (LLDs) with W2V2 transformer embeddings. The W2V2 toolkit [23] is based on a convolutional neural network encoder, a contextualized transformer network and a quantization module. This neural network model [21] was developed for letter recognition and has as input a speech signal. The core of W2V2 is the contextualized transformer, which takes as input the encoder feature vectors and processes it through 12 Transformer layers for the Base version of the model, or 24 layers for the Large version. The size of each layer varies from 768 (Base version) to 1 024 (Large version). The W2V2 transformer embeddings have already been used in other speech-related applications, such as emotion recognition [24], depression detection [25], speech impairment detection [26], and recently stuttered speech detection [15, 27].

We used for the Challenge the Base and Large version models. The two models are pre-trained on 960 hours of labeled speech from the LibriSpeech corpus [23]. The W2V2 toolkit extracts a 768-dimensional (1024-dimensional for the Large version) feature vector from the speech signal every 20 ms for each of the 12 (24 for the Large version) given layers of the contextualized transformer. To obtain features from a given node of a given layer, functionals were applied on successive embedding values of the node. For the W2V2 basic system, we have chosen 21 functionals [Table 2] (6 spatial parametric functionals, 10 spatial non-parametric functionals, 5 temporal parametric functionals). Each layer yields a different representation that might be more or less suitable for a task than a later or previous layer. This is the reason why we did not choose layers in a first time. For all 12 layers, the total number of features extracted for each speech chunk is equal to the number of layers multiplied by the layer size and multiplied by the number of functionals, i.e. $12 \times 768 \times 21 = 193,536$ features called W2V2 Basic audio-feature set. For the Large version model, the total number of features extracted for each speech chunk is equal to $24 \times 1024 \times 21 = 516,096$ features called W2V2 large audio-feature set. All calculations were performed in less than 24 hours using eight cores and 64 GB of memory. No Graphics Processing Unit (GPU) was used.

Table 2. Global Functionals

A) 6 Spatial parametric functionals
Mean (M), Standard Deviation (SD), Skewness (S), Kurtosis (K), HyperSkewness (HS), HyperKurtosis (HK)
B) 10 Spatial non-parametric functionals
Percentile 10% (P10), Percentile 25% (P25), Percentile 40% (P40), Percentile 50% (P50), Percentile 60% (P60), Percentile 75%, (P75), Percentile 90% (P90),
Percentile range 25–75 % (PR1), Percentile range 40–60 % (PR2), Percentile range 10–90 % (PR3)
C) 5 Temporal parametric functionals
Linear regression slope (L), Linear regression approximation Error (LE), Quadratic regression coefficient a (QA), Quadratic regression coefficient b (QB), Quadratic regression approximation Error (QE)

For the development of the Stuttered speech detector, we have chosen for the W2V2 Embeddings Basic System (W2BS) a linear Kernel SVM classifier with the W2V2 Basic audio-feature set. Scikit-learn toolbox [28] was used for the implementation of the SVM classifier. Complexity parameters of the SVM classifier were optimized to maximize the Unweighted Average Recall (UAR) on the Devel set. The Stutter Speech detector gave an UAR of 43% on the Devel set of KSF-C, an improvement of 12.8% compared to the baseline performance of the Challenge (30.2%).

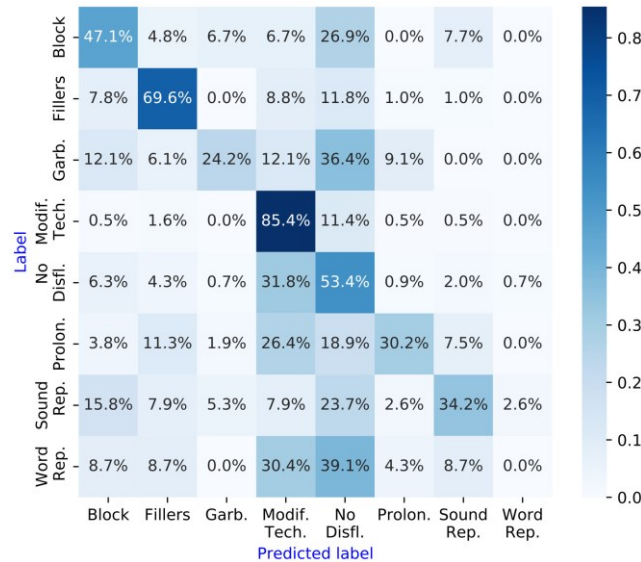


Figure 1: Confusion Matrix of the Stutter Speech detector on the Devel set for eight classes: Block, Fillers, Garbage, Modified speech techniques, No disfluency, Prolongation, Sound Repetition and Word/Phrase repetition.

Comparing the confusion matrix of the W2V2-embeddings Basic system [Figure 1] with the baseline confusion matrix [20], we notice an improvement for six classes: Block (47.1% vs 31.7%), Fillers (69.6% vs 24.5%), Prolongation (30.2% vs 13.2%), Modified speech techniques (85.4% vs 74.6%), Sound Repetition (34.2% vs 26.3%) and No disfluency (53.4% vs 12.4%) and a drop for two classes: Garbage (24.2% vs 54.5%) and Word/Phrase repetition (0% vs 4.3%).

4 W2V2-EMBEDDING ADVANCED SYSTEM

The W2V2 Basic/core audio feature set (193,536 features) performed well, but certainly many features are redundant or non-discriminative. It is not possible to study the audio-features one by one. We chose to group the audio-features by layer, then by layer and functional to determine the most discriminating layers and functionals.

4.1 Discriminating layers

The W2V2 Basic audio-feature set was splitted into twelve audio-feature sets called S_i by grouping the audio-features corresponding to the same layer i . Each audio-feature set S_i includes 16,128 audio-features. Figure 2 gives the results in UAR of the Stutter Speech detector for the twelve audio-feature sets S_i . We can notice that the maximum is reached for the audio-feature sets corresponding to layers 5 and 6 of the W2V2 transformer. The minimum is obtained for layer 12. This could indicate that the stuttering speech detection task is quite different from the speech recognition task for which the transformer was trained.

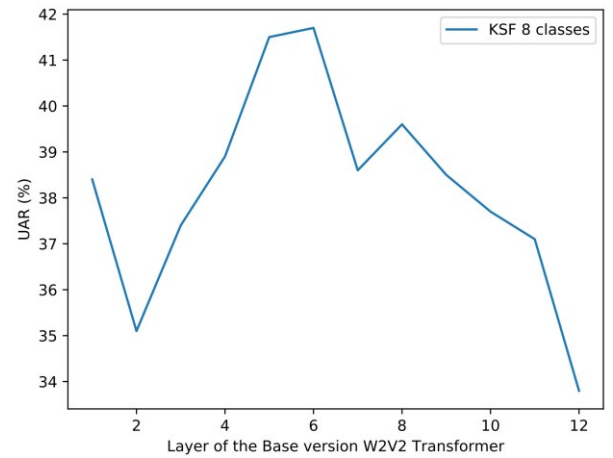


Figure 2: UAR results on the Devel set of Stutter speech detector for audio-feature sets S_i corresponding to layer i .

The results on the Devel set of KSF-C using the audio-features set S_6 were an UAR of 41.7% which is an improvement of 11.0 % compared to the official baseline performance of the Challenge (30.2%). It should be noted that the result on the Devel set is almost the same (41.7% vs 43.0%) as that obtained by the W2V2 Basic audio-feature set which includes 12 times more audio-features.

4.2 Discriminating layers and functionals

The W2V2 Basic audio-features set has been splitted into 252 audio-feature sets called S_{IJ} by grouping the audio-features corresponding to the same layer I and the same functional J . Each audio-feature set S_{IJ} includes 768 audio-features. Figure 3 gives the UAR of the Stutter Speech detector for the 252 audio-feature sets S_{IJ} per functional and per layer. We can notice that the most discriminating audio-feature sets correspond to the following 13 functionals: Mean, Standard Deviation, Percentiles (7), Percentile ranges (2) and regression approximation errors (2). An advanced audio-feature set called W2V2 Advanced audio-features set (9,984 features) has been defined from 13 audio-feature sets, the most discriminating by functional: S_{8M} , S_{5SD} , S_{5P10} , S_{6P25} , S_{5P40} , S_{8P50} , S_{6P60} , S_{5P75} , S_{5P90} , S_{5PR1} , S_{5PR2} , S_{5LE} , S_{5QE} . The Stutter Speech detector with the W2V2 Advanced audio-features set gave an UAR of 45.2% on the Devel set of KSF-C.

5 W2V2-EMBEDDING LARGE ADVANCED SYSTEM

The W2V2 Large audio-features set (516, 096 features) has been splitted into 504 audio-feature sets called L_{IJ} by grouping the audio-features corresponding to the same layer I and the same functional J . Each audio-feature set L_{IJ} includes 1024 audio-features. A large advanced audio-feature set called W2V2 Large Advanced audio-features set (15,360 features) has been defined from 15 audio-feature sets, the most discriminating by functional: L_{17M} , L_{17SD} , L_{19S} , L_{7P10} , L_{9P25} , L_{9P40} , L_{9P50} , L_{17P60} , L_{17P75} , L_{10P90} , L_{17PR1} , L_{19PR2} , L_{22PR3} , L_{17LE} , L_{17QE} . The Stutter Speech detector with the W2V2 Large Advanced audio-features set gave an UAR of 49.0% on the Devel set of KSF-C.

6 EXPERIMENTS ON THE TEST SET

The official baseline on the Test set is 40.4% in terms of UAR. Table 3 gives the UAR results of the Stutter Speech detector on the Test set for various audio features sets.

Table 3. Experiments on the Test set

#sub	audio feature set	#features	UAR
2	W2V2 basic (S_6)	16,128	48.1%
3	W2V2 basic	193,536	48.2%
4	W2V2 advanced	9,984	49.7%
5	W2V2 large advanced	15,360	52.3%

The best result, 52.3%, is achieved by the large advanced W2V2 audio feature set (15,360 features). This is a significant improvement of 11.9% over the official baseline.

7 CONCLUSION AND FUTURE WORKS

In this paper, we defined audio features from W2V2 transformer embeddings using 21 various functionals to obtain equal-sized feature vectors for each speech utterance. We studied the UAR results of the stuttered speech detector on the Devel set to determine the most discriminative audio-features by grouping them by layer and functional. The most sensible layers to the speech stuttering were the layers 5 and 6 on the 12 layers of Base version of W2V2 transformer and the layers 17 and 9 on the 24 layers of Large version. The most discriminating functions are in order of UAR results were percentiles, mean, regression approximation errors, percentile ranges and standard deviation. Two audio features sets specializing in stuttering speech have been developed and assessed. Experiments have shown an improvement of 9.3% to the first one and 11.9% for the second one on the Test set compared to the official baseline performance of the Challenge (40.4%). For the future works, we plan to introduce new functionals to better model the temporal evolution, fine-tune [29, 30] the W2V2 network with stuttering speech and analyze the changes to the transformer embeddings in terms of UAR results.

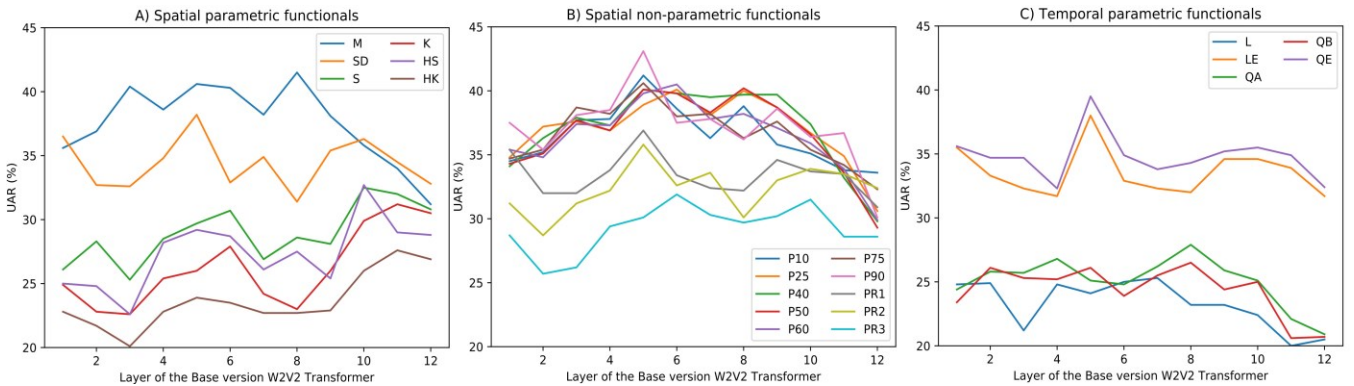


Figure 3: Results in UAR on the Devel set of the Stutter Speech detector for the audio-feature sets S_{IJ} (with I the layer and J the global functional [Table 2])

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